Solution:

The four year moving averages are shown in the last column.

Year	Annual value	4-year moving total	Centered total	4-year moving average
1998	154.0			
		-		
1999	140.5		-	
		590.0		
2000	147.0		1168.9	146.11
		578.9		
2001	148.5		1159.4	144.93
		580.5		
2002	142.9		1150.6	143.83
		570.1		
2003	142.1		1134.4	141.8
		564.3		
2004	136.6		1131.4	141.43
		567.1		
2005	142.7		1137.2	142.15
		570.1		
2006	145.7		1141.4	142.68
		571.3		
2007	145.1		-	
		-		
2008	137.8			

7.3.4 Method of least squares

Among the four components of the time series, secular trend represents the long term direction of the series. One way of finding the trend values with the help of mathematical technique is the method of least squares. This method is most widely used in practice and in this method the sum of squares of deviations of the actual and computed values is least and hence the line obtained by this method is known as the line of best fit.

It helps for forecasting the future values. It plays an important role in finding the trend values of economic and business time series data.

Computation of Trend using Method of Least squares

Method of least squares is a device for finding the equation which best fits a given set of observations.

Suppose we are given n pairs of observations and it is required to fit a straight line to these data. The general equation of the straight line is:

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